Supplemental Material: Empirical Methodology

Contents

This document describes the construction of advertiser value and externality scores from the XNP400 dataset, the three auction mechanisms used in the empirical simulations, the genetic algorithm used to optimize the threshold function, and the expected penalty curves derived from the optimized threshold. Notation follows Tables **??** and **??** in the main text.

1 Dataset and Score Construction

1.1 XNP400 Dataset

We use the XNP400 dataset, a corpus of $N = 11{,}853$ posts (after filtering) collected from the X (Twitter) platform as of February 2, 2025. For each post i we observe:

Symbol	Source field	Description
η_i	`impression_count`	Total lifetime impressions
α_i	`action_count`	Total engagements (likes, retweets, replies, quotes, bookmarks)
μ_i	derived	Age in months: (Feb 2, 2025 − created_at$_i$)/30

Each post is also associated with a set of Community Notes $\mathcal{J}(i)$. Note $j \in \mathcal{J}(i)$ carries a binary direction $\pi_{ij} \in \{-1, +1\}$ and receives a set of helpfulness ratings $\mathcal{L}(i, j)$ from the community.

1.2 Data Collection and Filtering

Posts were retrieved from the X Community Notes public data release as of February 2, 2025. We restrict the sample to posts that satisfy all of the following criteria:

1. **At least one Community Note.** We include only posts that attracted at least one note. Posts without any Community Note carry no externality signal and are excluded.
2. **Positive impressions.** Posts with zero recorded lifetime impressions are dropped, as their per-month scores are undefined.
3. **Outlier trimming.** We remove the top and bottom 0.1% of posts by aggregate post rating β_i and by total action count α_i. This trims approximately 24 posts at each tail and prevents extreme outliers from distorting the score distributions.
4. **Minimum agreement score.** Posts with an agreement score below −400 (rating units per 1,000 impressions) are removed. These posts have extraordinarily negative community assessments that are likely the result of viral coordinated flagging rather than genuine advertiser-relevant externalities.

After applying all filters the sample contains $N = 11{,}853$ posts spanning posts created from the platform's early period through early 2025.

1.3 Community Notes Scoring

Each rating $l \in \mathcal{L}(i, j)$ is mapped to a helpfulness score:

$$\rho_{ijl} = \begin{cases} 1 & \texttt{helpfulnessLevel} = \texttt{HELPFUL} \\ 0.5 & \texttt{helpfulnessLevel} = \texttt{SOMEWHAT\_HELPFUL} \\ 0 & \texttt{helpfulnessLevel} = \texttt{NOT\_HELPFUL} \end{cases} \tag{1}$$

The note direction encodes whether the note author classified the post as misleading:

$$\pi_{ij} = \begin{cases} -1 & \texttt{classification} = \texttt{MISINFORMED\_OR\_POTENTIALLY\_MISLEADING} \\ +1 & \texttt{classification} = \texttt{NOT\_MISLEADING} \end{cases} \tag{2}$$

The aggregate rating of post i sums the signed, weighted contributions of all notes and their ratings:

$$\beta_i = \sum_{j \in \mathcal{J}(i)} \pi_{ij} \Big(1 + \sum_{l \in \mathcal{L}(i,j)} \rho_{ijl}\Big), \tag{3}$$

where the additive 1 inside the parentheses treats the note author as an implicit rater who agrees with their own note. Posts with no notes have $\beta_i = 0$. A post flagged by many raters as misleading has $\beta_i \ll 0$; one rated as helpful has $\beta_i > 0$.

1.4 Value and Externality Scores

Both scores are expressed as monthly rates to control for post age, following the E1 and V1 methods described in the main text.

Externality score (E1)

The per-post externality (dollars per month) is

$$e_i = \zeta \cdot \frac{\beta_i}{\mu_i}, \tag{4}$$

where $\zeta \geq 0$ is the *externality cost* parameter (dollars per unit of aggregate rating per month). Because most posts that attract Community Notes are flagged as misleading, $\beta_i < 0$ for the majority of posts in the sample, so $e_i < 0$ at any positive ζ.

Advertiser value score (V1)

The per-post advertiser value (actions per month) is

$$v_i = \theta \cdot \frac{\alpha_i}{\mu_i}, \tag{5}$$

where $\theta \geq 0$ is the *action cost* parameter (dollars per engagement action per month). We set $\theta = 1$ throughout, so v_i is measured directly in actions per month and serves as the advertiser's declared value in the auction.

1.5 Calibration of ζ

We calibrate the externality cost ζ using two published estimates of the dollar value of social media content externalities.

Impression-based estimate (Goldstein et al.). Taking a CPM (cost per thousand impressions) of \$1.53, the implied cost per unit of aggregate rating is

$$\zeta^{\text{imp}} = \frac{1.53/1000}{\overline{\beta_i/\eta_i}} \approx \$2.71 \text{ per rating unit per month}, \tag{6}$$

where $\overline{\beta_i/\eta_i}$ is the sample mean of aggregate rating per impression.

Action-based estimate (Acquisti et al.). Taking a willingness-to-pay of \$0.54 per engagement action, the implied cost is

$$\zeta^{\text{act}} = \frac{0.54}{\overline{\beta_i/\alpha_i}} \approx \$2.54 \text{ per rating unit per month}, \tag{7}$$

where $\overline{\beta_i/\alpha_i}$ is the sample mean of aggregate rating per action.

Both methods yield $\zeta \approx \$2.5$, consistent with each other. Because there is inherent uncertainty in these estimates, we treat ζ as a free parameter and report results across the range $\zeta \in [0.01, 100]$.

2 Auction Simulation

2.1 Setup

For each experimental cell (k, ζ, D, n) — allocation slots, externality cost, polynomial degree, and bidders per draw — we maintain two independent sets of auction draws with different purposes.

Training draws ($J_{\mathbf{train}} = 500$). These are used exclusively by the genetic algorithm as its fitness signal during optimization. The training seed is a function of (ζ, n) only — it excludes k and D — so all degrees and slot counts within the same (ζ, n) cell train on identical post samples. This ensures that any difference in optimized welfare across degrees or k values reflects the effect of those parameters and not sampling variation in the training data.

Evaluation draws ($J_{\mathbf{eval}} = 2000$). These are held out from the optimizer entirely and used only for final welfare reporting. The evaluation seed is offset from the training seed by a fixed constant, guaranteeing independence between the two sets. Like the training seed, the evaluation seed excludes k and D, so welfare numbers are directly comparable across those dimensions for any fixed (ζ, n). The larger draw count (2000 vs. 500) reduces Monte Carlo variance in the reported welfare estimates.

Each draw j is formed by sampling n posts without replacement from the dataset. Post i enters draw j with pair (v_i, e_i) computed via (5) and (4).

2.2 Threshold Function

The collateralized mechanisms use a polynomial threshold of degree D:

$$\tau(e; \mathbf{c}) = \sum_{d=0}^{D} c_d \, e^d, \qquad \mathbf{c} = (c_0, c_1, \ldots, c_D) \in \mathbb{R}^{D+1}. \tag{8}$$

The coefficient vector $\mathbf{c}$ is the decision variable optimized by the genetic algorithm (Section 3).

2.3 Three Auction Mechanisms

2.3.1 VCG (Counterfactual)

All n advertisers participate. The k advertisers with the highest values are selected:

$$\mathcal{S}_j^{\mathrm{vcg}} = \operatorname*{arg\,max}_{S \subseteq [n],\, |S| = k} \sum_{i \in S} v_i. \tag{9}$$

Social welfare in draw j is

$$W_j^{\mathrm{vcg}} = \sum_{i \in \mathcal{S}_j^{\mathrm{vcg}}} (v_i + e_i). \tag{10}$$

2.3.2 Participant Audit (VCGA)

Only advertisers whose declared value clears the threshold are admitted:

$$\mathcal{M}_j(\mathbf{c}) = \big\{ \, i \in [n] : v_i \geq \tau(e_i; \mathbf{c}) \big\}. \tag{11}$$

Among admitted advertisers, the top k by value are selected:

$$\mathcal{S}_j^{\text{vcgPA}}(\mathbf{c}) = \underset{S\subseteq\mathcal{M}_j,\,|S|\le\min(k,\,|\mathcal{M}_j|)}{\arg\max} \sum_{i\in S} v_i. \tag{12}$$

If fewer than k advertisers are admitted, all admitted advertisers are selected. Social welfare is

$$W_j^{\text{vcgPA}}(\mathbf{c}) = \sum_{i\in\mathcal{S}_j^{\text{vcgPA}}} (v_i + e_i). \tag{13}$$

The auction payment (generalised $(k{+}1)$th-price rule among admitted bidders) is

$$p_j^{\text{PA}}(\mathbf{c}) = \begin{cases} v_{(k+1)}^{\mathcal{M}} & |\mathcal{M}_j| \ge k+1 \\ 0 & \text{otherwise,} \end{cases} \tag{14}$$

where $v_{(k+1)}^{\mathcal{M}}$ is the $(k{+}1)$th highest value among the admitted set.

2.3.3 Recipient Audit (RA)

No participation threshold is applied. Each advertiser's individual social value is $w_i = v_i + e_i$. The k advertisers with the largest *positive* social value are selected:

$$\mathcal{S}_j^{\text{vcgRA}} = \underset{S\subseteq\{\,i:w_i>0\,\},\,|S|\le k}{\arg\max} \sum_{i\in S} w_i. \tag{15}$$

If fewer than k advertisers have positive social value, all such advertisers are selected. Social welfare is

$$W_j^{\text{vcgRA}} = \sum_{i\in\mathcal{S}_j^{\text{vcgRA}}} w_i. \tag{16}$$

2.4 Welfare Aggregation

All reported welfare numbers use the $J_{\text{eval}} = 2000$ evaluation draws. Mean welfare for each mechanism is

$$\bar{W}^{\text{vcg}} = \frac{1}{J_{\text{eval}}} \sum_{j=1}^{J_{\text{eval}}} W_j^{\text{vcg}}, \quad \bar{W}^{\text{vcgPA}}(\mathbf{c}^*) = \frac{1}{J_{\text{eval}}} \sum_{j=1}^{J_{\text{eval}}} W_j^{\text{vcgPA}}(\mathbf{c}^*), \quad \bar{W}^{\text{vcgRA}} = \frac{1}{J_{\text{eval}}} \sum_{j=1}^{J_{\text{eval}}} W_j^{\text{vcgRA}}, \tag{17}$$

where $\mathbf{c}^*$ is the optimal threshold found by the genetic algorithm. Welfare gains relative to the VCG counterfactual are

$$\Delta W^{\text{PA}} = \bar{W}^{\text{vcgPA}}(\mathbf{c}^*) - \bar{W}^{\text{vcg}}, \qquad \Delta W^{\text{RA}} = \bar{W}^{\text{vcgRA}} - \bar{W}^{\text{vcg}}. \tag{18}$$

Both quantities are reported in dollars per month per auction draw.

3 Genetic Algorithm Optimization

The goal is to find

$$\mathbf{c}^* = \underset{\mathbf{c}\in\mathcal{C}}{\arg\max}\ \bar{W}^{\text{vcgPA}}(\mathbf{c}), \tag{19}$$

where $\mathcal{C}$ is the gene space defined below. Because the objective is non-convex in $\mathbf{c}$ and the feasible space is continuous and high-dimensional, we use a genetic algorithm (GA) operating directly in original (v, e) space without any normalization.

3.1 Gene Space

The gene space $\mathcal{C} = \mathcal{C}_0 \times \mathcal{C}_1 \times \cdots \times \mathcal{C}_D$ is derived from the empirical distribution of (v_i, e_i) values so that the search range scales correctly regardless of the externality cost ζ or the polynomial degree D.

Let σ_v denote the standard deviation of advertiser values across all bidder draws, and let $\overline{|e|^d} = \mathbb{E}[|e_i|^d]$ denote the mean absolute dth power of externalities.

The intercept gene space ensures the threshold can start above or below all bids:

$$\mathcal{C}_0 = \left[v_{\min} - \tfrac{1}{2}\sigma_v,\ \ v_{\max} + \tfrac{1}{2}\sigma_v\right]. \tag{20}$$

For slope coefficients ($d \geq 1$), the range is calibrated so that at a typical externality value, the contribution $|c_d \cdot e^d|$ can shift τ by up to $\lambda\sigma_v$ — covering the full spread of advertiser values:

$$\mathcal{C}_d = \left[-\lambda\, \sigma_v / \overline{|e|^d},\ \ \lambda\, \sigma_v / \overline{|e|^d}\right], \quad d = 1, \ldots, D, \tag{21}$$

where $\lambda = 3$ is the range multiplier (Table 1).

3.2 Vectorized Fitness Evaluation

All N_{pop} candidate solutions in the current population are evaluated simultaneously using a single matrix multiply per auction draw. Let $\mathbf{P} \in \mathbb{R}^{N_{\text{pop}} \times (D+1)}$ be the population matrix (row g is candidate $\mathbf{c}^{(g)}$) and let $\mathbf{E}^{(j)} \in \mathbb{R}^{n \times (D+1)}$ be the Vandermonde-style power matrix for draw j with entries $E^{(j)}_{ip} = e_i^p$. Then

$$\mathbf{T}^{(j)} = \mathbf{P}\left(\mathbf{E}^{(j)}\right)^{\top} \in \mathbb{R}^{N_{\text{pop}} \times n}, \qquad T^{(j)}_{g,i} = \tau(e_i;\, \mathbf{c}^{(g)}), \tag{22}$$

evaluating the threshold for every population member and every advertiser at once. The admission mask is $A^{(j)}_{g,i} = \mathbf{1}[v_i \geq T^{(j)}_{g,i}]$. Advertisers are sorted by v descending; the cumulative sum of admitted indicators identifies the first k admitted positions, yielding the winner set. Welfare is accumulated by summing $v_i + e_i$ over winners for all g simultaneously. The fitness of candidate g is

$$F_g = \frac{1}{J_{\text{train}}} \sum_{j=1}^{J_{\text{train}}} W_j^{\text{vcgPA}}(\mathbf{c}^{(g)}), \tag{23}$$

accumulated across the $J_{\text{train}} = 500$ training draws without storing per-draw matrices. The evaluation draws are never seen by the optimizer.

3.3 Selection, Crossover, and Mutation

Selection. Tournament selection: for each parent slot, a random subset of the population is drawn without replacement and the candidate with the highest fitness wins.

Crossover. Single-point crossover: a random cut point in $\{1, \ldots, D\}$ is chosen for each pair of parents, and the offspring receives coefficients from parent A before the cut and from parent B after.

Mutation. Gaussian perturbation with per-gene step sizes proportional to each gene's range:

$$\sigma_d^{\mathrm{mut}} = \sigma_{\mathrm{frac}} \cdot |\mathcal{C}_d|, \tag{24}$$

where $|\mathcal{C}_d| = c_d^{\max} - c_d^{\min}$ is the gene range width and σ_{frac} is the mutation sigma fraction (Table 1). Each gene is independently perturbed with probability p_{mut}; to prevent stagnation, at least one gene per offspring is always mutated. All perturbed values are clipped to the gene bounds.

3.4 Multi-Restart and Warm Start

To increase coverage of the coefficient space, each experimental cell is run for R independent restarts, each with a different random seed. The solution with the highest fitness across all restarts is retained.

Warm start. When sweeping polynomial degrees $D = 1, 2, 3$ in ascending order, the optimal solution from degree D is used to seed the first population member for degree $D + 1$:

$$\mathbf{c}_{\mathrm{seed}}^{(D+1)} = \big(c_0^*, c_1^*, \ldots, c_D^*,\ 0\big). \tag{25}$$

The warm start is applied only to restart 0; remaining restarts use a fresh random population, preserving the benefit of the warm start while maintaining diversity.

Fair cross-degree comparison. All degrees within the same cell (k, ζ, n) use an identical set of J bidder draws (controlled by a shared sample seed that excludes the degree index). This ensures that welfare differences across degrees reflect the effect of polynomial flexibility rather than sampling variation.

3.5 Hyperparameters

Table 1: Genetic algorithm hyperparameters (defaults used unless noted otherwise).

Symbol	Default	Description
D	1, 2, 3	Polynomial degree of τ (swept)
N_{pop}	50	Population size
T	500	GA generations per restart
R	1	Independent restarts per cell
N_{mate}	15	Parents selected for crossover each generation
p_{mut}	0.25	Per-gene mutation probability
σ_{frac}	0.15	Gaussian mutation step as fraction of gene range
λ	3.0	Gene range multiplier (Eq. 21)
J_{train}	500	Training auction draws per cell (GA fitness)
J_{eval}	2000	Evaluation draws per cell (welfare reporting only)
n	20	Bidders sampled per auction draw

4 Expected Penalty Curves

Given the optimized threshold $\tau(e;\, \mathbf{c}^*)$, we compute the expected penalty an advertiser with externality value e would face under each audit mechanism.

4.1 Participant Audit: $R^{\mathbf{PA}}(e)$

Under the participant audit, an advertiser with externality e is excluded from any draw in which the market-clearing price p_j^{PA} falls below the threshold $\tau(e;\, \mathbf{c}^*)$. To clear the threshold in such a draw, the advertiser would need to post collateral equal to the shortfall $\tau(e;\, \mathbf{c}^*) - p_j^{\mathrm{PA}} > 0$.

Let $\{p_j\}_{j=1}^{J_{\mathrm{eval}}}$ be the sequence of auction payments from (14), computed on the evaluation draws. Define

$$A(e) = \frac{1}{J_{\mathrm{eval}}} \big| \{ j : p_j < \tau(e;\, \mathbf{c}^*) \} \big|, \tag{26}$$

$$P(e) = \mathbb{E}\big[p_j \;\big|\; p_j < \tau(e;\, \mathbf{c}^*) \big]\,. \tag{27}$$

Here $A(e)$ is the fraction of draws in which the market price is below the threshold, and $P(e)$ is the mean price conditional on being below the threshold. The expected participant penalty at externality e is

$$R^{\mathrm{PA}}(e) = A(e)\, \big[\tau(e;\, \mathbf{c}^*) - P(e) \big]. \tag{28}$$

For large negative externalities, the threshold $\tau(e; \mathbf{c}^*)$ is high relative to the market price, so both $A(e)$ and the expected gap are large, resulting in a steep penalty. For posts with small or positive externalities, $\tau(e; \mathbf{c}^*)$ is low and most market prices clear it, so $A(e) \approx 0$ and $R^{\mathrm{PA}}(e) \approx 0$.

4.2 Recipient Audit: $R^{\mathbf{RA}}(e)$

Under the recipient audit, selection is based on social value $w_i = v_i + e_i$. An advertiser with externality e internalises the full cost of its externality:

$$R^{\mathrm{RA}}(e) = -e. \tag{29}$$

This is linear in e: advertisers generating negative externalities ($e < 0$) pay a penalty equal to $|e|$, while those generating positive externalities ($e > 0$) effectively receive a subsidy.